

Wiring the Brain

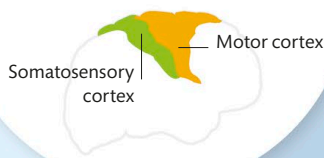
Until recently, only the brain controlled the body's muscles and glands. But next-generation electrical, mechanical, and robotic devices—often developed after limb loss—are extending its abilities.

Bionic limbs

Motorized bionic limbs now exist that react to activity in the brain's motor cortex, responding to instructions sent as tiny electrical impulses along motor nerves. These increasingly powerful prostheses can also provide sensory feedback so that the brain's control systems can provide delicate ongoing control, more closely mimicking the natural limb or other body part.

1 Motor cortex

The brain's movement center formulates patterns of motor nerve impulses that naturally coordinate dozens of muscles to move the arm and hand.



2 Sending impulses

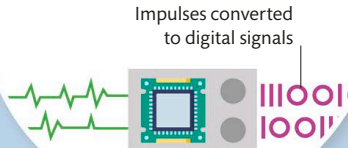
Motor nerve impulses travel from the brain via the spinal cord along peripheral nerves to the arm and hand.



Pattern of nerve activity

3 Microprocessor

Microchips change the nerve impulses into digital signals understood by the circuits and motors of the bionic part.



4 Bionic hand

Up to 10 servos (small, lightweight motors) drive movements of the hand and fingers, pivoting at self-sensing joints.



Hand receives processed signals and converts them to movement

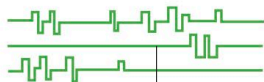
Wires carry digital signals to servos in hand

Two-way communication

The motor cortex masterminds movements of the bionic part. As with a natural limb, these are continually modified by interchange with the somatosensory cortex.

6 Mindful awareness

Further processing converts the sensory signals to more natural forms that can be interpreted by the brain's touch center, the somatosensory cortex.



Electrical pulses

5 Sensory data

Receptors in the hand's motors, joints, and artificial skin generate responses.



Feedback signals produced by robotic hand in digital form

Motor impulses to bionic hand

Sensory signals from bionic hand

Median, radial, and ulnar nerves